



Research article

A novel hybrid forecasting system for crude oil futures prices: A dual perspective of deterministic forecasting and uncertainty analysis

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ABSTRACT

The crude oil market occupies a crucial place in the composition of the global economy, however, the high level of uncertainty in crude oil prices makes it a challenging task to construct reliable forecasting techniques to obtain reliable expected results. Therefore, to effectively cope with the high uncertainty and the technical limitations of existing forecasting studies, this study constructs a hybrid forecasting system from the dual perspectives of deterministic forecasting and uncertainty analysis. With the concept of “denoising-integration” as the core, the system integrates several denoising techniques, econometric methods, and artificial intelligence methods, and establishes a hybrid forecasting model with high accuracy and robustness with the assistance of multi-objective optimization theory. Furthermore, by introducing fuzzy theory onto the basis of deterministic prediction results, an interval forecasting model with a narrower interval width and better comprehensive performance is constructed under the premise of ensuring that the coverage of prediction intervals meets the requirements of the confidence level. In this study, the forecasting system is applied to four global core crude oil futures markets, and the empirical study finds that the performance is improved by more than 50 % compared with the benchmark model, which has an impressive performance advantage.

1. Introduction

The immeasurable strategic position of crude oil (CO) as the major source of energy for global production makes its price volatility closely linked to macroeconomic dynamics [1]. Therefore, the view that detailed analysis and forecasting of CO prices is crucial for the development of formulation management plans, stabilization of economic and social development, and diversification of investment risks has become a consensus in the academic community [2]. However, given the swift fluctuations in geopolitical dynamics and market sentiments, the sequence studied is often non-stationary, irregular, and rough, and the performance of the existing forecasting technologies is facing serious challenges. Therefore, dependable and precise price forecasting research is the long-term theme of global scholars [3].

After a relatively comprehensive review of the extant literature in the field of crude oil price forecasting, existing research techniques can be categorized into three groups; econometric methods, machine learning (ML) and artificial intelligence (AI) techniques,

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and hybrid techniques [4].

Econometric methods are dominated by time series analysis and regression analysis, representative of which are autoregressive integrated moving average (ARIMA) and vector autoregression (VAR) [5], etc. This type of method has been widely used in the early stage of the research on CO price forecasting. Represented by Lutz Kilian and his co-authors, a monthly crude oil futures price forecasting model was built based on VAR, which proved that VAR possesses superior performance than autoregressive and autoregressive moving average in crude oil futures price forecasting [6]. Similarly, they built separate forecasting models based on VAR for different crude oil price influencing factors and beat the unchanged forecasting and random wandering models in terms of performance through a weighting strategy [7,8]. Furthermore, Mohammadi and Su (2010) modeled and forecasted the weekly CO spot price based on ARIMA [9]. However, such methods are restricted by strict prior assumptions, and their forecasting performance has been widely criticized due to their inability to effectively adapt to the nonlinear characteristics of (CO) prices [10]. With the advancement of knowledge and technology, ML and AI techniques that can capture the nonlinear and non-stationary characteristics of the series more effectively have gradually replaced the traditional econometric methods as cutting-edge technology in CO price forecasting by virtue of their flexibility, accuracy, and universal applicability [11]. For example, Fan et al. (2016) verified its superiority in CO research by building a support vector machine (SVM) model [12].

Although ML and AI techniques can significantly improve forecasting accuracy, a large body of evidence shows that such models generally suffer from poor interpretability, weak generalization, low robustness as well as applicability, and are prone to fall into the traps of overfitting and local optimality [13]. Therefore, in recent years, hybrid models coupling data preprocessing strategies, econometric methods, and artificial intelligence techniques, aiming to synthesize the advantages of each party to obtain more accurate prediction results, have received more attention [14].

Many studies have confirmed that daily frequency CO futures price series usually exhibit low signal-to-noise ratio, strong volatility, non-stationarity, heteroskedasticity, autocorrelation, and other complex features that hinder the effectiveness of statistical analysis and modeling. Therefore, a single model is not sufficient to accurately forecast the expected changes in CO prices. Currently, the integration model and the decomposition-integration model are the two hybrid forecasting models for CO prices that are most frequently employed [15]. Integrated model refers to combining the forecasts of multiple single models [16], and this type of approach has been shown to significantly outperform single forecasting models in CO price forecasting [17]; decomposition-integration model, on the other hand, is the approach that decomposes data into different subsequences, then models these subsequences independently, and ultimately integrates their forecasts [18], allowing for the effective capture of the multiple time scales and different features present in the data [19]. As in the research of Sun et al. (2022), more accurate forecasting results were obtained by decomposing the initial CO futures price series into a series of components [20].

However, existing studies on hybrid forecasting of CO prices suffer from two shortcomings in that on the one hand less attention is paid to the noise inherent in the crude oil price series, and on the other hand fewer researches include considerations of the uncertainty that exists in the forecasting process.

Distinguished from the integrated model and decomposition-integration model, the core of the construction of a denoising-based hybrid forecasting model is to remove the highest frequency noise component from the original sequence and train the model using reconstructed sequences to reduce the interference of noisy information in the modeling process. Wang et al. (2022) built a denoising-based hybrid model to achieve higher accuracy in forecasting the price of agricultural products [21]; Xu and Wang (2023) also claimed that the denoising technique can better tap the true trend of the sequence and significantly enhance the individual model's capacity [22]. Therefore, it is natural to assume that denoising is a key step to improve the quality of analyzing and forecasting daily frequency CO futures price series.

In addition, with the profound understanding of scholars on nonlinear complex systems, the results of deterministic point forecasts on account of expectations can no longer satisfy the demands of risk management in the modern CO market [23]. In the global environment of sharp fluctuations in CO prices [24], the interval forecasting method that grasps the uncertainty of expected CO price changes with the help of uncertainty analysis can provide more useful references for the users of forecasting information [25] and can minimize the risk of CO market management [26], which is one of the optimal means of efficiently avoiding the risk of forecasting in the present time and has received extensive attention from scholars [27]. Based on whether they are strictly constrained by a priori statistical assumptions, interval forecasting models can be divided into parametric and nonparametric models [28]. The former may result in forecasting failure in practical applications due to inaccurate assumptions of distribution. Also, although commonly used nonparametric models such as Bootstrap and upper and lower bound estimation can adaptively generate forecast intervals, they are not suitable for forecasting complex and variable CO futures prices due to the complex computational process and the risk of overfitting [29].

According to the above viewpoints, this study integrates the cutting-edge ideas of denoising, integration, and multi-objective optimization. First, the denoising module is integrated into the prediction system, aiming at fully mining the real trend and changing the rule of the sequence, reducing the misleading nature of the noise, and intensifying the comprehensive performance. However, the traditional denoising algorithms have poor generalization ability and poor adaptability to different noise types and other problems, and it is difficult to make the most reasonable denoising algorithm choice in practice, therefore, this study introduces four common denoising algorithms in the denoising module which has been proved to be excellent in many empirical evidences, to make up for the shortcomings of a single denoising algorithm. Second, nine forecasting techniques, including econometric models as well as shallow and deep machine learning models, are introduced into the forecasting module, and the performance advantages of each single model are combined with the assistance of multi-objective optimization theory, which significantly improves its forecasting accuracy and robustness. Finally, this study introduces the fuzzy theory and establishes a fuzzy interval prediction model with better interval width and comprehensive performance while ensuring that the coverage ratio meets the risk level requirements. The empirical study shows

that the hybrid forecasting system can provide accurate and robust CO price deterministic forecasting and uncertainty analysis results for the four core CO futures markets around the world, which effectively expands the reference sources for risk assessment by information users.

The contribution of this study lies in the following points.

1. **The introduction of diversified denoising techniques significantly reduces the interference of uncertainty in data noise on the forecasting results.** The method can effectively synthesize the technical advantages of multiple denoising techniques, providing a reliable data basis for accurate and reliable forecasting of CO futures prices.
2. **With the concept of “denoising-integration” as the core, the hybrid point prediction model that combines the performance advantages of multiple denoising techniques, econometric models, and multiple shallow or deep neural network models is constructed,** which combines both the advantages of prediction performance and robustness. This technology breaks through the limitations of the traditional single data denoising and prediction techniques and can show high performance, high robustness, and high efficiency in different scenarios.
3. **Introducing the theory of fuzzy clustering, the fuzzy interval prediction model is constructed on the premise of ensuring that the data points in each category follow a similar distribution function.** The method effectively avoids the failure of the full sample distribution assumption due to the sudden change of data features and has significant performance and efficiency advantages.

The organizational framework of this study can be summarized into four main parts: the second part offers a full summary of the study framework and the technical principles of the hybrid research system on account of the dual view of deterministic forecasting and uncertainty analysis proposed in this study. The third part describes the research dataset of interest, the construction of evaluation metrics, and the empirical research findings. The fourth part then summarizes the study.

2. The research framework of the dual-view hybrid forecasting system

This section describes in detail the research framework of the proposed dual-view hybrid forecasting system for CO futures. Fig. 1 graphically depicts the specific workflows of this research framework and details the principles of the core technologies involved in each module.

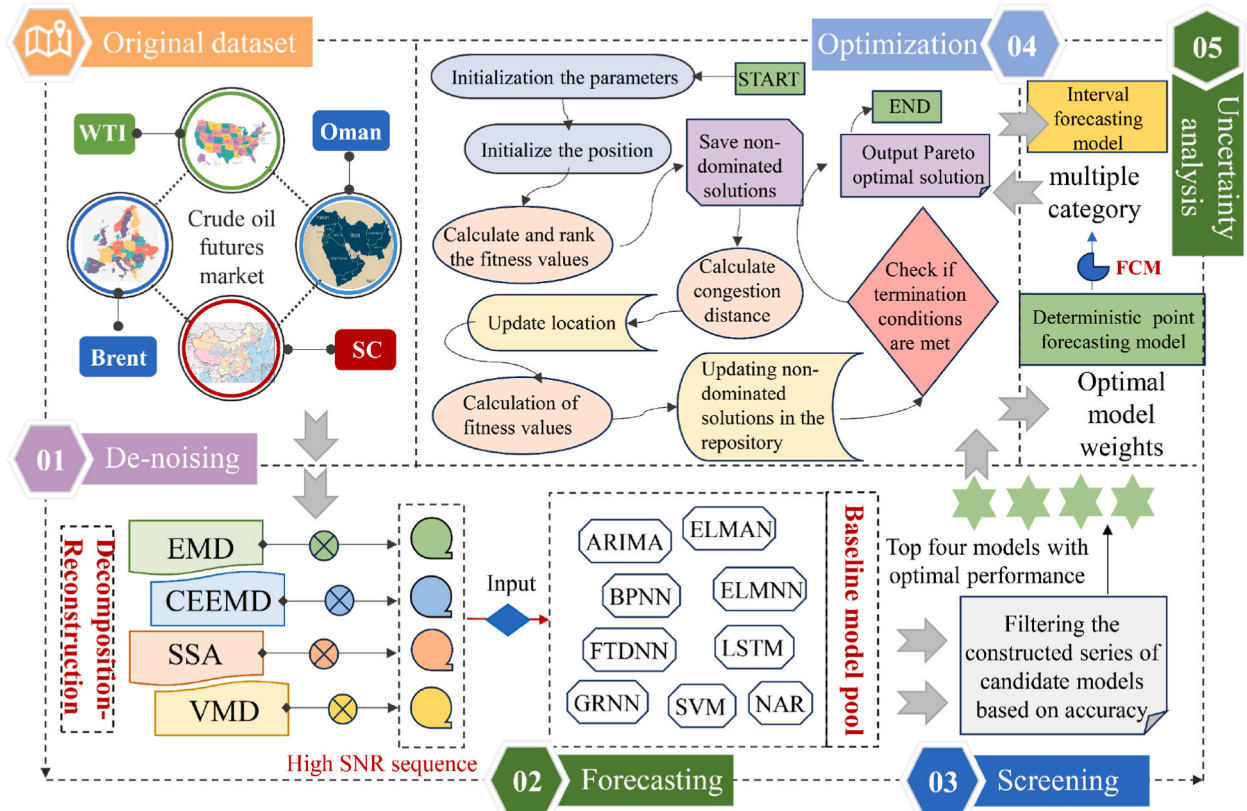


Fig. 1. Flowchart of the research framework.

2.1. Module 1: de-noising

Four de-noising techniques have been introduced in this module to eliminate high-frequency noise components in the original series $A^0(t)$ and obtain reconstructed sequences $A^{de}(t) = \sum_{i=1}^{n-1} IMF_i(t)$ with high signal-to-noise ratios, which are Empirical Mode Decomposition (**EMD**) [30], Complementary Ensemble Empirical Mode Decomposition (**CEEMD**) [31], Singular Spectrum Analysis (**SSA**) [32], and Variational Mode Decomposition (**VMD**) [33]. The principles of these four techniques are shown below:

- 1) **EMD**: For the time series $A^0(t)$, curve interpolation is introduced to fit its extreme points to obtain the upper and lower envelopes $\tilde{A}^0(t)$ and $A^0_{\sim}(t)$ of the series, and $A^0_{m-1}(t) = (\tilde{A}^0(t) + A^0_{\sim}(t))/2$ is obtained by averaging. If the difference $D^0_{m-1}(t)$ between $A^0(t)$ and $A^0_{m-1}(t)$ is a smooth sequence, then $D^0_{m-1}(t) = IMF_1^{EMD}(t)$ is the Intrinsic Mode Function (**IMF**) of $A^0(t)$. Otherwise, repeat the above steps until the condition in Equation (1) holds.

$$\sum_{t=0}^T \frac{|D^0_{m,k-1}(t) - D^0_{m,k}(t)|^2}{D^0_{m,k-1}(t)^2} < \tau, \quad (1)$$

and then $IMF_1^{EMD}(t)$ is obtained, where $0.2 < \tau < 0.3$ is the threshold value.

Accordingly, $R_1^{EMD}(t) = A^0(t) - IMF_1^{EMD}(t)$ is the first-order residual sequence. Continuing to repeat the above steps n times by using the corresponding residual sequences $R_i^{EMD}(t)$, $i = 1, 2, \dots, n-1$ as the sequences to be decomposed, $IMF_i^A(t)$, $i = 2, 3, \dots, n$ can be obtained such that $A^0(t) = \sum_{i=1}^n IMF_i^{EMD}(t) + R_n^{EMD}(t)$.

- 2) **CEEMD**: The algorithm is grounded in the **EMD** algorithm, and aims at eliminating the inherent modal aliasing and end effect problems. For each original sequence $A^0(t)$ (or the i -th order residual sequence $R_i^{EMD}(t)$) to be decomposed, pairs of opposite positive and negative white noises $(-1)^k \varepsilon \theta^j(t)$ are added as auxiliary noises to generate M new sequences $A^0(t) + (-1)^k \varepsilon \theta^j(t)$ (or $R_i^{EMD}(t) + (-1)^k \varepsilon \theta^j(t)$) to be decomposed, after which the principle of **EMD** is applied to each new sequence to be decomposed to obtain the corresponding M modal components $[imf_j]_{j=1}^M$, and then the corresponding **IMF** is $IMF_i^{CEEMD} = \sum_{j=1}^M imf_j$.

- 3) **SSA**: The algorithm consists of four key steps:

Embedding. This step involves choosing a suitable window length L and performing a hysteresis operation on $A^0(t) = [A^0(1), A^0(2), \dots, A^0(T)]$ to obtain the trajectory matrix $\vec{G}_{L \times K}$ presented in Equation (2), habitually, $L < N/2$, $\vec{G}_{L \times K}$ and $K = T - L + 1$.

$$\vec{G}_{L \times K} = \begin{bmatrix} A^0(1) & A^0(2) & \dots & A^0(K) \\ A^0(2) & A^0(3) & \dots & A^0(K+1) \\ \dots & \dots & \dots & \dots \\ A^0(L) & A^0(L+1) & \dots & A^0(N) \end{bmatrix} \quad (2)$$

Decomposing. Using the singular value decomposition operation can decompose $\vec{G}_{L \times K}$ into $\vec{G}_{L \times K} = \vec{U}_{\vec{G}} \Sigma_{\vec{G}} \vec{V}_{\vec{G}}^T$, where $\vec{U}_{\vec{G}}$ and $\vec{V}_{\vec{G}}$ are the unit orthogonal matrix satisfying $\vec{U}_{\vec{G}} \vec{U}_{\vec{G}}^T = I$ and $\vec{V}_{\vec{G}} \vec{V}_{\vec{G}}^T = I$. $\Sigma_{\vec{G}}$ is a diagonal matrix with singular values on the diagonal.

Grouping. The L components of $\vec{G}_{L \times K}$ are divided into n disjoint groups, which $[\vec{G}_{I1}, \vec{G}_{I2}, \dots, \vec{G}_{In}]$ is the decomposition result of the time series $A^0(t)$.

- 4) **VMD**: The solution process of its **IMF** can be summarized as a constrained variational problem as shown in Equation (3):

$$\min_{\{IMF_i^{VMD}\}, \{\omega_i\}} \left\{ \sum_i \|\partial_t [\Psi(t) \otimes IMF_i^{VMD}(t)] e^{-j\omega_i t}\|_2^2 \right\} \text{ s.t. } \sum_i IMF_i^{VMD}(t) = A^0(t), \quad (3)$$

where $\Psi(t) = \delta(t) + j/(\pi t)$, $\otimes$ represents the convolution operation, $\delta(t)$ is the unit pulse function, and ω_i is the center frequency. After that, the alternating direction multiplication method is introduced to find the corresponding $IMF_i^{VMD}(t)$.

2.2. Module 2: forecasting

This module introduces **ARIMA**, Backpropagation Neural Network (**BPNN**) [34], Elman Neural Network (**ELMAN**) [35], General Regression Neural Network (**GRNN**) [36], Extreme Learning Machine (**ELMNN**) [37], **SVM** [38], Focused Time-Delay Neural Network (**FTDNN**) [39], Nonlinear Autoregressive (**NAR**) [40], and Long Short-Term Memory (**LSTM**) [41] models as baseline models and trains the sub-forecasting models by using the raw unprocessed sequences $A^0(t)$ and the reconstructed sequences $A^{de}(t)$ with the noise component removed, obtained from **Module 1**, as the model inputs, respectively.

2.3. Module 3: screening

To avoid excessive computational burden due to too many models, this module screens out the four sub-models with the highest accuracy from a series of constructed sub-models based on the performance evaluation results as the candidate models for the combined model, whose corresponding out-of-sample forecast values are $\hat{P}^j(t)$, $j = 1, 2, 3, 4$.

2.4. Module 4: optimization

In this module, the Multiobjective Slime Mold Algorithm (**MOSMA**) is chosen to optimize the weighting weights of the sub-models, and the optimization objectives and constraints are shown in Equation (4):

$$\min \begin{cases} \frac{1}{T} \sum_{t=1}^T \left| \frac{A^0(t) - \sum_{j=1}^4 \omega^j \hat{P}^j(t)}{A^0(t)} \right| \times 100\% \\ \frac{1}{T} \sum_{t=1}^T \left| A^0(t) - \sum_{j=1}^4 \omega^j \hat{P}^j(t) \right| \end{cases} \quad (4)$$

subject to : $\sum_{j=1}^4 \omega^j = 1$

MOSMA is derived from the Slime Mold Algorithm (**SMA**) [42], which is a metaheuristic population-based optimization algorithm that follows the logic of the oscillatory behavior of slime molds during experiments, which simulates the behavioral patterns of slime molds in grasping, entangling, and approaching food [43], and the movement process can be simplified as Equation (5).

$$\vec{P}_s(iter+1) = \begin{cases} rand \cdot (upper - lower) + lower, & rand < 0.03 \\ \vec{P}_s^{op}(iter) + \vec{arg}_{vb} \cdot \left(\vec{w} \cdot \vec{P}_s^A(iter) - \vec{P}_s^B(iter) \right), & r < \varepsilon, \\ \vec{arg}_{vc} \cdot \vec{P}_s(iter), & r \geq \varepsilon \end{cases} \quad (5)$$

where $[upper, lower]$ is the search boundary, $r \in [0, 1]$ and $rand \in [0, 1]$ varies random. $\vec{arg}_{vb} \in [-\arctanh(-(\text{iter}/\text{max_iter}) + 1), \arctanh(-(\text{iter}/\text{max_iter}) + 1)]$, iter and max_iter are the present and maximum number of iterations, $\vec{arg}_{vc}$ varying linearly from 1 to 0. $\vec{P}_s$ is the location of the slime mold in the population, $\vec{P}_s^A$ and $\vec{P}_s^B$ are two random locations, and $\vec{P}_s^{op}$ is the location of the individual with the highest fitness value. $\vec{w}$ is the weight vector as shown in Equation (6). Moreover, $\varepsilon = \tanh|Fit^m - Fit^{op}|$, and Fit^m stands for the fitness value of the m -th individual, $m = 1, 2, \dots, M$, Fit^{op} is the favorite fitness value searched so far.

Table 1
Pseudo-code for the **MOSMA** algorithm.

<i>Algorithm: MOSMA</i>	
1.	<i>/*Setting the internal parameters. */</i>
2.	<i>/*Initially generating a random population of slime mold individuals $\vec{P}_s^{0,m}$, $m = 1, 2, \dots, M$. */</i>
3.	<i>/*Measuring fitness of the individuals $Fit^{0,m}$, $m = 1, 2, \dots, M$. */</i>
4.	<i>/*Compute the rank of non-dominated solutions and obtain the first frontier. */</i>
5.	<i>/*Preserving non-dominated solutions in the Pareto frontier. */</i>
6.	<i>/*Compute congestion distance. */</i>
7.	$iter = 1$
8.	WHILE $iter \leq \text{max_iter}$
9.	FOR EACH $\vec{P}_s^{0,m}$, $m = 1, 2, \dots, M$
10.	<i>/* Updating the position of $\vec{P}_s^{0,m}$, $\vec{P}_s^{update,m}$, $m = 1, 2, \dots, M$ */</i>
11.	IF $r < \varepsilon$
12.	$\vec{P}_s^{update,m}(iter+1) = \vec{P}_s^{op}(iter) + \vec{arg}_{vb} \cdot \left(\vec{w} \cdot \vec{P}_s^A(iter) - \vec{P}_s^B(iter) \right)$
13.	ELSE
14.	$\vec{P}_s^{update,m}(iter+1) = \vec{arg}_{vc} \cdot \vec{P}_s^{0,m}(iter)$
15.	END IF
16.	<i>/*Merge $\vec{P}_s^{0,m}$ and $\vec{P}_s^{update,m}$ such that $\vec{P}_s^{new,m} = \vec{P}_s^{0,m} \cup \vec{P}_s^{update,m}$. */</i>
17.	<i>/*Measuring the fitness value of the new individuals $\vec{P}_s^{new,m}$. */</i>
18.	<i>/*Sorting and calculating congestion distances. */</i>
19.	<i>/*Updating the saved non-dominated solutions. */</i>
20.	END FOR
21.	$iter = iter + 1$
22.	END WHILE

$$\vec{w} = \begin{cases} 1 + r \cdot \log\left(\frac{Fit^{op-c} - Fit^m}{Fit^{op-c} - Fit^{wo-c}} + 1\right), & Fit^m \text{ ranks first half of the population} \\ 1 - r \cdot \log\left(\frac{Fit^{op-c} - Fit^m}{Fit^{op-c} - Fit^{wo-c}} + 1\right), & \text{others} \end{cases}, \quad (6)$$

where $r \in [0, 1]$ is random, Fit^{op-c} and Fit^{wo-c} are the optimal and worst fitness values in the present stage.

Following on from **SMA**, **MOSMA** introduces a Pareto-optimal mechanism (**POM**) for obtaining non-dominated solutions (**NDSs**), incorporates an elitist non-dominated ordering method to compute the non-dominated ordering of all non-dominated solutions, obtains the first frontier, and uses a congestion distance operator to ensure increased solution diversity. The corresponding pseudo-code is shown in [Table 1](#).

2.5. Module 5: uncertainty analysis

In this module, the deterministic forecasting results are firstly clustered into C categories based on the fuzzy C-mean (**FCM**) clustering method, and it is assumed that each category possesses a similar data structure. After that, **MOSMA** is implemented to optimize the forecasts of the points $\hat{P}(t) = \sum_{j=1}^4 \omega^j \hat{P}^j(t)$ in each category and generate forecast intervals with a 95 % confidence level, its optimization objective and constraints are as indicated in Equation (7):

$$\min \begin{cases} \frac{1}{T} \sum_{t=1}^T \kappa(t) - 0.95, & \kappa(t) = 1, \text{ if } L(t) \leq A^0(t) \leq U(t) \\ & \kappa(t) = 0, \text{ else} \\ \frac{1}{T} \sum_{t=1}^T [U(t) - L(t)] \end{cases} \quad (7)$$

$$\text{subject to : } \begin{cases} \frac{1}{T} \sum_{t=1}^T [U(t) - L(t)] > 0 \\ 0.95 \leq \frac{1}{T} \sum_{t=1}^T \kappa(t) \leq 1 \end{cases}$$

After obtaining the forecast intervals of each category, the forecast intervals of all the categories are combined to obtain the final forecast interval $[L(t), U(t)]$. **FCM** is a machine learning technique that is extensively applied to data analysis and pattern recognition, and can provide more flexible clustering results by incorporating fuzzy theory, the principle of which can be briefly described as follows:

Step 1: Initialize the affiliation matrix. The affiliation degree φ_t^c portrays the affiliation relationship between the point forecasting results $\hat{P}(t)$, ($t = 1, 2, \dots, T$) and the clustering center v_c ($c = 1, 2, \dots, C$), and can generalize the specific concept of $\{0, 1\}$ (i.e., $\{\text{not belonging/belonging}\}$) in the traditional clustering algorithm to the fuzzy concept of categories within the (0,1) interval, and the initial affiliation degree matrix is the $\Phi_{t \times c}^{(0)} = [\varphi_t^c]$. The larger φ_t^c the closer the data object is to the clustering center.

Step 2: Determine if the maximum number (max_iter) of iterations has been exceeded. If exceeded, the fuzzy clustering result is output and the current iteration is launched. If not exceeded, continue with subsequent operations (max_iter is specified in advance).

Step 3: Compute the clustering center. The c -th clustering center is calculated as Equation (8).

$$v_c = \frac{\sum_{t=1}^T (\varphi_t^c \cdot \hat{P}(t))}{\sum_{t=1}^T \varphi_t^c} = \sum_{t=1}^T \left(\frac{\varphi_t^c}{\sum_{t=1}^T \varphi_t^c} \cdot \hat{P}(t) \right). \quad (8)$$

Step 4: Updating the affiliation matrix as Equation (9).

$$\Phi_{t \times c}^{(iter)} = [\varphi_t^{c-iter}], \varphi_t^{c-iter} = \frac{1}{\sum_{k=1}^C \left(\frac{\|\hat{P}(t) - v_c\|}{\|\hat{P}(t) - v_k\|} \right)^{\frac{2}{C-1}}} \quad (9)$$

Step 5: Determine if the iteration termination condition $\max_{t,c} \left\{ |\varphi_t^{c-iter} - \varphi_t^{c-iter-1}| \right\} \leq \varepsilon$ is satisfied. ε is the error threshold. This iteration condition implies that the global/local optimum is considered to have been reached when the affiliation does not change significantly, and the iteration ends, returning the fuzzy clustering results.

3. Experiments and evaluations

In this section, three empirical experiments are carried out to confirm the superior performance of the model proposed in this study, and the process of implementing the experiments, the conclusions, and also the data sets and evaluation metrics required are described in detail.

3.1. Data sets

In this study, the daily closing prices of four global core CO futures contracts, including the West Texas Intermediate CO futures contract (**WTI**) listed and traded on the New York Mercantile Exchange (NYMEX), Brent CO futures contract (**Brent**) listed, and traded on the Intercontinental Exchange (ICE), Oman CO futures contract (**Oman**) launched on the Dubai Mercantile Exchange (DME), and Shanghai CO futures contract (**SC**) traded on the Shanghai International Energy Exchange (SIEX), are selected as the research object. The data were obtained from the WIND database as well as the website of Investing¹ and each contains 1288 daily frequency data points.

The dataset samples are divided into two parts, where the training set contains 1000 data points and the testing set contains 288 data points. Fig. 2 showcases the variation of the trend of the dataset over the study interval, while Table 2 showcases the basic statistical information of the training and test sets, respectively. From Fig. 2 and Table 2, it's clear that the daily frequency CO price series exhibits extreme volatility, non-stationarity, and non-normality over the sample interval.

In line with the modeling process of the hybrid forecasting system proposed in this study, we first apply the denoising procedure to the original research sequences for pre-processing so as to reduce the noise interference in the sequences and provide more reliable data support for the subsequent empirical modeling. Fig. 3 presents the comparison between the denoised sequences applying various denoising procedures and the original sequences, from which it can be found that the denoising practices can significantly diminish the noise components of the sequences, and the performance of denoising is not the same for different denoising practices.

3.2. Evaluation metrics

In this study, the evaluation metrics system is constructed from two perspectives: point forecasting and interval forecasting, respectively.

3.2.1. Construction of evaluation metrics system for point forecasting models

Accuracy is the crux for evaluating point forecasting models, therefore, this study constructs the evaluation metrics system of point forecasting models from three dimensions, namely, error accuracy, trend accuracy, and directional accuracy.

For the point forecast value $\hat{P}_t$ and the true value A_t , $t = 1, \dots, T$. Error accuracy portrays the overall degree of deviation between all forecast values in the out-of-sample forecast and the true value, and contains four main indicators as shown in Equations (10)–(13):

$$\text{Mean absolute percentage error (MAPE : \%)} = \frac{1}{T} \sum_{t=1}^T \left| \frac{A_t - \hat{P}_t}{A_t} \right| \times 100\% ; \quad (10)$$

$$\text{Mean absolute error (MAE)} = \frac{1}{T} \sum_{t=1}^T |A_t - \hat{P}_t| ; \quad (11)$$

$$\text{Mean square error (MSE)} = \frac{1}{T} \sum_{t=1}^T (A_t - \hat{P}_t)^2 ; \quad (12)$$

$$\text{Root mean square error (RMSE)} = \sqrt{\frac{1}{T} \sum_{t=1}^T (A_t - \hat{P}_t)^2} . \quad (13)$$

Trend accuracy describes how well the forecasting model fits the real-time series data of CO prices and is referred to by the metric in Equation (14):

$$\text{R squared (R}^2\text{)} = 1 - \frac{\sum_{t=1}^T (A_t - \hat{P}_t)^2}{\sum_{t=1}^T (\bar{A}_t - A_t)^2}, \quad (14)$$

where $\bar{A}_t = (1/T) \sum_{t=1}^T A_t$.

Directional accuracy reflects the consistency of the direction of change of all out-of-sample predicted values with the true values, and measures the accuracy of the model's forecasts at the level of upward and downward trends in CO futures prices, as shown in Equation (15):

¹ Data acquisition website: <https://www.wind.com.cn/portal/zh/EDB/index.html> and <https://cn.investing.com/>.

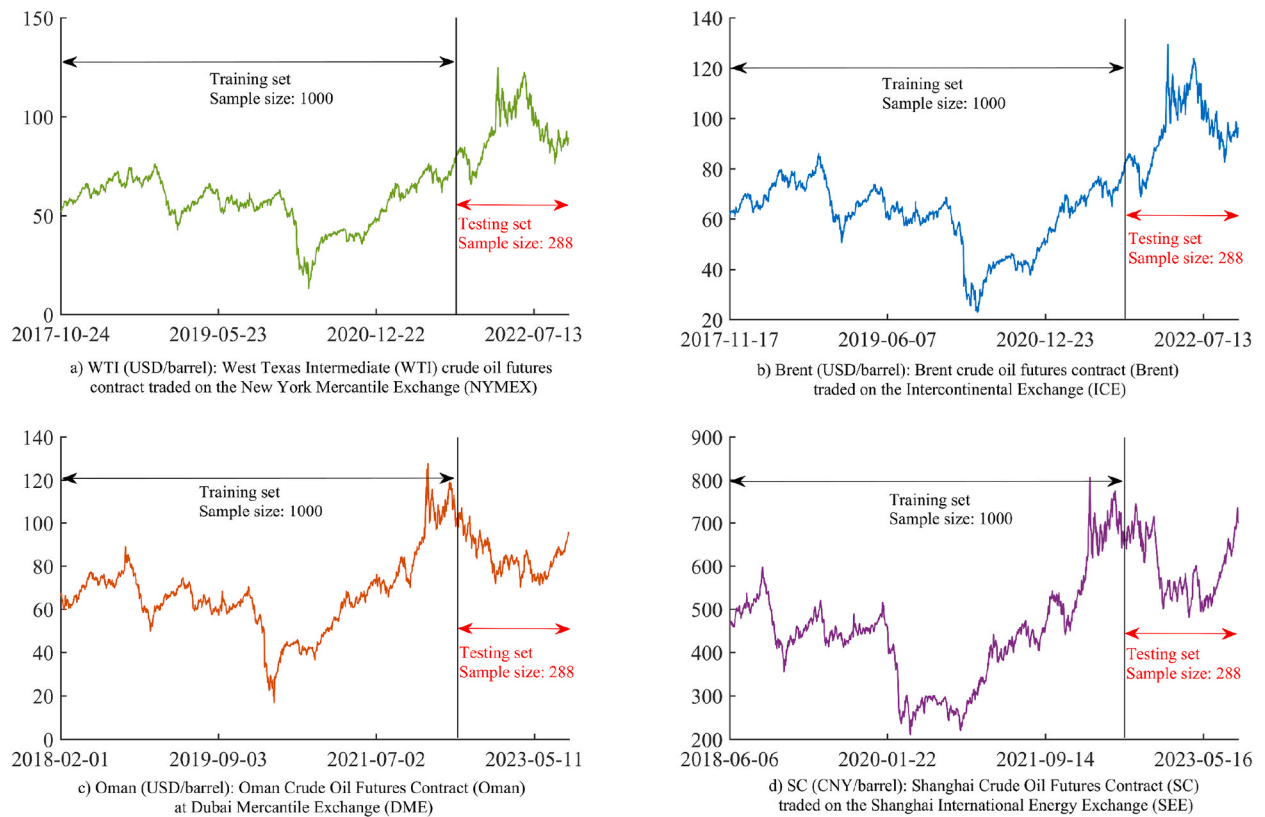


Fig. 2. Chart of the original research dataset.

Table 2

Basic statistical information for each series in the dataset.

		Minimum	Maximum	Mean	Std	Stability	Kurtosis	Skewness	Normality
WTI	Training set	16.82	127.71	66.3319	17.8059	N	4.1378	0.4990	N
	Testing set	70.35	107.10	84.8372	8.2454	N	2.5244	0.4897	N
Brent	Training set	23.00	86.10	61.4059	12.9948	N	3.1062	−0.8591	N
	Testing set	68.94	129.47	96.0438	12.9737	N	2.2871	0.1324	N
Oman	Training set	13.12	79.08	56.2726	12.2888	N	3.5629	−0.8974	N
	Testing set	65.71	124.77	92.5240	13.1843	N	2.3527	0.3085	N
SC	Training set	210.60	806.60	442.9956	118.1394	N	3.2732	0.4313	N
	Testing set	481.50	745.20	597.7594	68.4382	N	1.7778	0.3074	N

Note: “N” signifies that the corresponding series is not smooth (or normal), while “Y” signifies that the corresponding series is smooth (or normal).

WTI: in USD per barrel; Brent: in USD per barrel; Oman: in USD per barrel; SC: in CNY per barrel.

$$\text{Directional accuracy (DACC)} = \frac{1}{T-1} \sum_{t=2}^T \mathbf{1}(\text{sgn}(\hat{P}_t - A_{t-1}) = \text{sgn}(A_t - A_{t-1})) \times 100\%, \quad (15)$$

where $\text{sgn}(\cdot)$ is the sign function and $\mathbf{1}(\cdot)$ is the indicator function.

3.2.2. Construction of evaluation metrics system for interval forecasting models

Interval prediction results always appear in pairs in the form of $[L_t, U_t]$, $t = 1, \dots, T$. Therefore, this study first measures the probability that the true observation A_t falls within the upper and lower bounds of the interval prediction to evaluate the reliability. That is, if $L_t \leq A_t \leq U_t$, the true observation A_t is considered to fall in the prediction interval $[L_t, U_t]$ with $\kappa_t = 1$, and vice versa with $\kappa_t = 0$. The corresponding evaluation metrics are shown in Equations (16)–(19):

$$\text{Forecast interval coverage probability (FICP)} = \frac{1}{T} \sum_{t=1}^T \kappa_t \times 100\%. \quad (16)$$

However, excessive pursuit of high FICP may lead to the interval width being too wide and the interval forecasting results are not

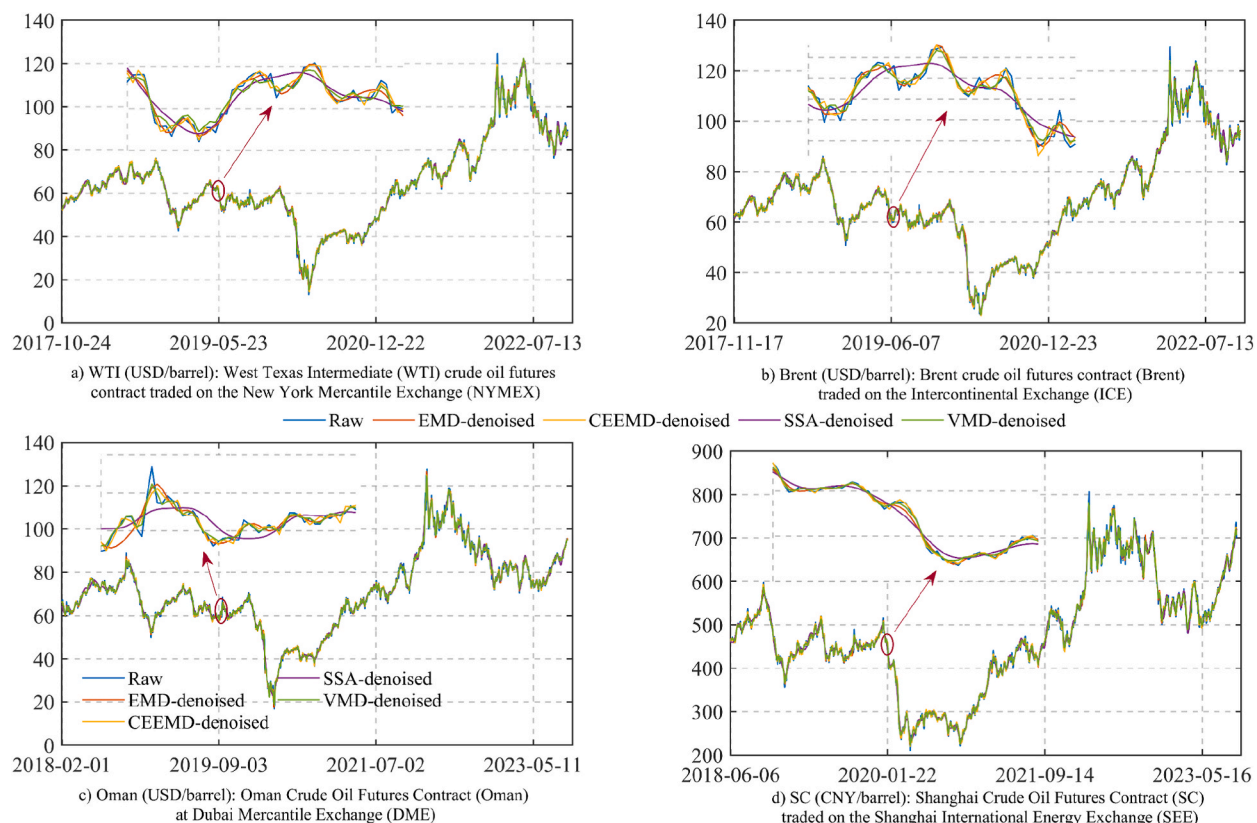


Fig. 3. Sequence diagram after various denoising procedures.

Table 3

Parameter list of the algorithms.

Algorithm	Parameter	Connotation	Value	Algorithm	Parameter	Connotation	Value
EMD	$[T_1, T_2, Tol]$	Vector of stopping parameters	[0.05, 0.5, 0.05]	SVM	TF	Transfer function	rbf
	It_{max}	Maximum number of sifting iterations	2000		n_{input}	Same as BPNN	8
CEEMD	N_m	Number of modes	10	LSTM	n_{output}		1
	N_{std}	Noise standard deviation	0.2		n_{input}	Same as BPNN	8
	N_r	Number of realizations	50		n_{output}		1
	N_m	Number of modes	10		EP_s	Number of epochs	200
	max_iter	Maximum number of sifting iterations	500		I	Initial learning rate	0.004
SSA	L_{wd}	Window length	[1, 50]	ELMAN	LP	Learning rate drop period	130
	N_m	Number of modes	10		LF	Learning rate drop factor	0.18
VMD	α	Moderate bandwidth constraint	2000	MOSMA	n_{input}	Same as BPNN	8
	τ	Noise-tolerance	0		n_{hidden}		10
	N_m	Number of modes	10		n_{output}		1
	tol	Tolerance of convergence criterion	1e-6		Pop	Size of population	100
BPNN	n_{input}	Number of input layer nodes	8	ARIMA	dim	Number of decision variables	4
	n_{hidden}	Number of hidden layer nodes	10		Nm	Size of the dominant population	50
	n_{output}	Number of output layer nodes	1		$Nobj$	Number of objectives	2
	n_{goal}	Goal of the train	0.0004		$[u, d]$	Random range	[0, 1]
GRNN	EP_s	Number of epochs	100	ELMNN	max_iter	Maximum number of iterations	300
	n_{input}	Same as BPNN	8		p	Auto-Regressive term	5
	n_{output}		1		d	Integrated term	1
	n_{spread}	Spread of the net	0.09		q	Moving average term	3
FTDNN/ NAR	n_{input}	Same as BPNN	8	ELMNN	n_{input}	Same as BPNN	8
	n_{hidden}		12		n_{hidden}		20
	n_{output}		1		n_{output}		1

informative, therefore, this study further measures the interval width and the corresponding metric is:

$$\text{Forecast interval normalized average width (FINAW)} = \frac{1}{TR} \sum_{t=1}^T [U_t - L_t]. \quad (17)$$

$R^* = A_t^{\max} - A_t^{\min}$ denotes the standardized treatment process. However, High FICP usually implies wider FINAW, which conflicts with interval prediction evaluation, prompting this study to search for more effective metrics for synthesizing evaluations.

$$\text{Coverage width - based criterion (CWC)} = \text{FINAW} (1 + \gamma e^{-\eta(\text{FICP}-\alpha)}), \quad (18)$$

where α is the predefined confidence level, if $\text{FICP} \geq \alpha$, $\gamma = 0$, otherwise, $\gamma = 1$. η is the penalty parameter.

$$\text{Average interval score (AIS)} = \frac{1}{T} \sum_{t=1}^T \xi_t, \xi_t = \begin{cases} -2(1-\alpha)I_t - 4(L_t - A_t), & A_t < L_t \\ -2(1-\alpha)I_t, & A_t \in [L_t, U_t] \\ -2(1-\alpha)I_t - 4(A_t - U_t), & A_t > U_t \end{cases}, \quad (19)$$

where $I_t = U_t - L_t$.

3.3. Empirical analysis

As to confirm the superiority of the hybrid system proposed in this study, three comparative experiments are carried out for all four datasets, where the first 1000 data points of the four datasets serve for fitting the models and the remaining 288 data points serve for evaluating the out-of-sample performance of the models. All empirical studies in this section are built under the same parameter settings, specifically, the specific parameter settings are shown in Table 3.

Experiment I: The main thrust of this experiment is to compare the out-of-sample performance of the hybrid forecasting system proposed in this study (**PROPOSED-CM**) with a set of classical time series forecasting models across all datasets. The classical time series models selected as benchmark models include **ARIMA** (parameters: $[p, d, q] = [5, 1, 3]$), **BPNN** (parameters: $[n_{\text{input}}, n_{\text{hidden}}, n_{\text{output}}, n_{\text{goal}}, EP_s] = [8, 10, 1, 0.0004, 100]$), **ELMAN** (parameters: $[n_{\text{input}}, n_{\text{hidden}}, n_{\text{output}}] = [8, 10, 1]$), **GRNN** (parameters: $[n_{\text{input}}, n_{\text{output}}, n_{\text{spread}}] = [8, 1, 0.09]$), **ELMNN** (parameters: $[n_{\text{input}}, n_{\text{hidden}}, n_{\text{output}}] = [8, 20, 1]$), **SVM** (parameters: $[TF, n_{\text{input}}, n_{\text{output}}] = [\text{rbf}, 8, 1]$), **FTDNN** (parameters: $[n_{\text{input}}, n_{\text{hidden}}, n_{\text{output}}] = [8, 12, 1]$), **NAR** (parameters: $[n_{\text{input}}, n_{\text{hidden}}, n_{\text{output}}] = [8, 12, 1]$), and **LSTM** (parameters: $[n_{\text{input}}, n_{\text{output}}, EP_s, I, LP, LF] = [8, 1, 200, 0.004, 130, 0.18]$). Numerically, the results of the one-step and two-step-ahead performance comparisons for the four datasets are shown in Fig. 4 and Tables 4 and 5, respectively, and it's evident that: in all four datasets, the one-step and two-step-ahead predictions of the **PROPOSED-CM** have significant performance improvement advantages, achieving the lowest error accuracy,

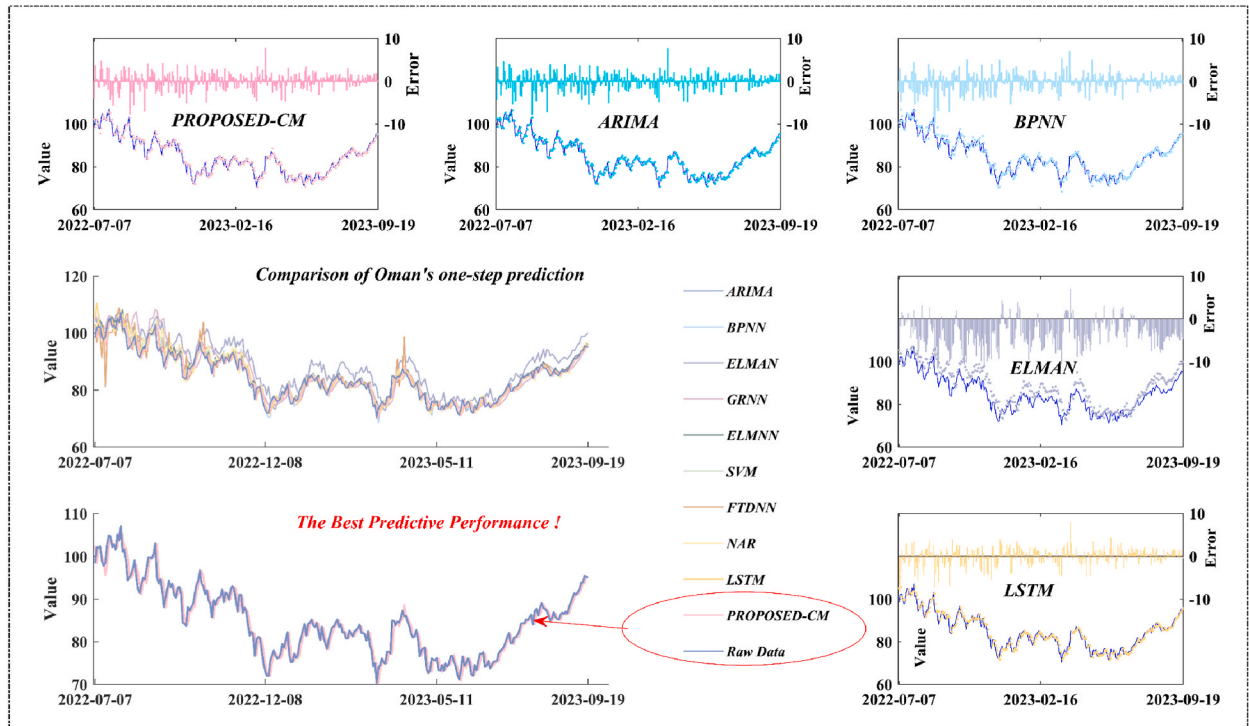


Fig. 4. Schematic diagram of Oman's one-step-ahead performance comparison.

the highest trend accuracy, and the highest direction accuracy compared to the benchmark model.

For **WTI**, in the one-step-ahead forecast, the four benchmark models with the best performance in terms of error accuracy and trend accuracy are **SVM** ($\text{MAPE}_{1\text{-step}}^{\text{SVM}} = 2.4447\%$; $\text{R}^2_{1\text{-step}}^{\text{SVM}} = 0.9265$), **ELMNN** ($\text{MAPE}_{1\text{-step}}^{\text{ELMNN}} = 2.5096\%$; $\text{R}^2_{1\text{-step}}^{\text{ELMNN}} = 0.9196$), **BPNN** ($\text{MAPE}_{1\text{-step}}^{\text{BPNN}} = 2.6996\%$; $\text{R}^2_{1\text{-step}}^{\text{BPNN}} = 0.9066$), and **LSTM** ($\text{MAPE}_{1\text{-step}}^{\text{LSTM}} = 2.7162\%$; $\text{R}^2_{1\text{-step}}^{\text{LSTM}} = 0.9063$). However, in two-step-ahead prediction, the top four benchmark models are **SVM** with $\text{MAPE}_{2\text{-step}}^{\text{SVM}} = 3.3382\%$, **LSTM** with $\text{MAPE}_{2\text{-step}}^{\text{LSTM}} = 3.3677\%$, **ARIMA** with $\text{MAPE}_{2\text{-step}}^{\text{ARIMA}} = 4.3848\%$, and **ELMNN** with $\text{MAPE}_{2\text{-step}}^{\text{ELMNN}} = 6.4043\%$, which show significant differences in performance compared to the benchmark models in one-step-ahead prediction.

At the same time, even in the same one-step-ahead forecasting, there are significant differences in the top-performing benchmark models in different research datasets. For example, compared with the ranking results of the benchmark models in **WTI**, the top four benchmark models in **Brent** are **ARIMA**, **SVM**, **LSTM**, and **NAR**, while in **Oman** and **SC**, they are **ARIMA**, **SVM**, **ELMNN**, and **BPNN**.

Therefore, the establishment of the hybrid forecasting model can not only synthesize the performance advantages of all sub-models to achieve comprehensive performance optimization but also improve the adaptability and robustness of the model to prevent decision-making errors due to model uncertainty.

Numerically, the proposed hybrid forecasting model (**PROPOSED-CM**) exhibits significant performance advantages over the benchmark model for all datasets and all performance metrics. Taking **Oman**'s one-step-ahead prediction as an example, the values of the performance metrics are $\text{MAPE} = 1.0761\%$, $\text{MAE} = 1.0133$, $\text{RMSE} = 1.3733$, $\text{MSE} = 1.8860$, $\text{R}^2 = 0.9891$, $\text{DACC} = 85.0174\%$, and the performance improvement percentages are in the ranges (50 %, 83 %), (50 %, 83 %), (50 %, 81 %), (76 %, 97 %), (4 %, 57 %), (53 %, 77 %).

Experiment II: The main thrust of this experiment is to compare the out-of-sample performance of hybrid forecasting systems based on different denoising algorithms for all four datasets. The hybrid prediction models used for comparison with the hybrid prediction model (**PROPOSED-CM**) proposed in this study include **RAW-CM**, **EMD-CM**, **CEEMD-CM**, **SSA-CM**, and **VMD-CM**, which are based on the original data, **EMD**-based denoised data (parameters: $[T_1, T_2, \text{ToI}] = [0.05, 0.5, 0.05]$, $\text{It}_{\text{max}} = 2000$, $N_m = 10$), **CEEMD**-based denoised data (parameters: $N_{\text{std}} = 0.2$, $N_r = 50$, $N_m = 10$, $\text{max_iter} = 500$), **SSA**-based denoised data (parameters: $L_{\text{wd}} = [1, 50]$, $N_m = 10$), and **VMD**-based data (parameters: $\alpha = 2000$, $\text{tau} = 0$, $N_m = 10$, $\text{tol} = 1e - 6$), respectively. Furthermore, the parameters of **MOSMA** used to optimize the weights of the sub-models are set to $\text{Pop} = 100$, $\text{dim} = 4$, $N_m = 50$, $N_{\text{obj}} = 2$, $[u, d] = [0, 1]$, $\text{max_iter} = 300$. The results of the out-of-sample performance evaluation metrics are displayed in [Table 6](#), from which we can draw the following conclusions:

It is easy to derive that there is a significant difference in the performance of the comparison models in different datasets as well as in different prediction steps. We first compare the same datasets: in the empirical study of **WTI**, in one-step-ahead forecasting, the best-performing comparison model is **VMD-CM** ($\text{MAPE} = 1.1638\%$), while in two-step-ahead forecasting, the optimal comparison model becomes **SSA-CM** ($\text{MAPE} = 2.6656\%$). Similarly, we study the same forecasting steps: in two-step-ahead forecasting, **SSA-CM** ($\text{MAPE} = 2.6656\%$) is the optimal comparison model corresponding to **WTI**, **Brent** corresponds to **SSA-CM** and **VMD-CM** ($\text{MAPE} = 2.8941\%$) and **Oman** corresponds to **VMD-CM** ($\text{MAPE} = 1.2346\%$), while **SC** corresponds to **EMD-CM** ($\text{MAPE} = 1.6545\%$). Therefore, the comprehensive consideration of different denoising techniques is an important prerequisite to improve the robustness of the model.

To be relative, the proposed hybrid prediction model achieves excellent performance benefits in multi-step prediction for all datasets. In terms of one-step-ahead forecasting, **PROPOSED-CM** significantly outperforms the other comparison models in all performance metrics in all three datasets of interest except **SC**. Also, in **SC**, although the hybrid model with the best performance is **VMD-CM**, its value of the error accuracy metric **MAPE** differs from that of **PROPOSED-CM** by less than 0.01 %. Moreover, in two-step-ahead prediction, **PROPOSED-CM** achieves optimality in all performance metrics, except for the special case of sample **WTI** in **DACC** performance.

In summary, the proposed hybrid model **PROPOSED-CM** significantly outperforms hybrid models based on other single denoising algorithms and its superior performance is robust and adaptable.

Experiment III: This experiment is designed to compare the forecasting performance of fuzzy interval prediction models based on different denoising algorithms. The fuzzy interval forecasting models involved, **RAW-CM-FIP**, **EMD-CM-FIP**, **CEEMD-CM-FIP**, **SSA-CM-FIP**, **VMD-CM-FIP**, and **PROPOSED-CM-FIP**, are based on the forecasting results of hybrid point forecasting models integrating different denoising techniques, i.e., **RAW-CM**, **EMD-CM**, **CEEMD-CM**, **SSA-CM**, **VMD-CM**, and **PROPOSED-CM**, respectively. The empirical comparison results are shown in [Table 7](#) and [Fig. 5](#), from which we can draw the following conclusions:

Overall, all of the fuzzy interval forecasting models modeled for the comparative analysis fulfilled the confidence level threshold required by the study, i.e., $\text{FICP} > 95\%$, which means that all of these fuzzy interval forecasting models are capable of providing an information base that meets the needs of management at a given level of risk in practice.

Specifically, in one-step-ahead forecasting, none of the proposed fuzzy interval prediction models (**PROPOSED-CM-FIP**) achieved optimality in terms of **FICP** in out-of-sample interval forecasting for all samples. However, in out-of-sample forecasting for all datasets except **Oman**, **PROPOSED-CM-FIP** significantly reduces **FINAW** and performs optimally on the composite metrics **AIS** and **CWC** compared to the other fuzzy interval forecasting models available for comparison. Taking the out-of-sample interval forecasting results for **WTI** as an example, compared with other fuzzy interval forecasting models for comparison, **PROPOSED-CM-FIP** improves the performance metric **FINAW** by 14 %–60 %, **AIS** by 12 %–60 %, and **CWC** by 14 %–60 %.

In two-step-ahead forecasting, **PROPOSED-CM-FIP** achieves optimality in all performance metrics in out-of-sample forecasting for **Oman** and **SC**, specifically, $\text{FICP}_{\text{PROPOSED-CM-FIP}}^{\text{Oman}} = 96.5157\%$, $\text{FINAW}_{\text{PROPOSED-CM-FIP}}^{\text{Oman}} = 0.1626$, $\text{AIS}_{\text{PROPOSED-CM-FIP}}^{\text{Oman}} = -0.6832$,

Table 4
Out-of-sample performance of the classical benchmark model versus the proposed hybrid model in **WTI** and **Brent**.

		Step-1						Step-2					
		MAPE	MAE	RMSE	MSE	R ²	DACC	MAPE	MAE	RMSE	MSE	R ²	DACC
WTI	ARIMA	3.6372	21.6037	26.8400	720.3851	0.8457	44.4056	4.3848	4.1335	5.5872	31.2171	0.8197	56.2937
	BPNN	2.6996	15.9455	20.8864	436.2430	0.9066	50.3497	12.6332	12.8226	16.6090	275.8578	−0.5930	45.8042
	ELMAN	6.5166	39.2724	48.8561	2386.9226	0.4888	47.2028	12.6051	12.7392	16.2790	265.0068	−0.5303	45.4545
	GRNN	3.9779	23.6585	30.0419	902.5186	0.8067	49.3007	14.7485	14.9014	18.8721	356.1543	−1.0567	45.8042
	ELMNN	2.5096	14.8414	19.3718	375.2665	0.9196	50.6993	6.4043	6.5072	9.1109	83.0078	0.5207	47.2028
	SVM	2.4447	14.5506	18.5252	343.1812	0.9265	47.5524	3.3382	3.1591	4.2751	18.2762	0.8945	51.3986
	FTDNN	3.4821	21.7136	52.0427	2708.4404	0.4199	49.3007	7.3209	7.4064	9.9455	98.9138	0.4288	45.8042
	NAR	3.0978	18.5393	23.3689	546.1058	0.8830	52.0979	14.2079	14.3008	18.0914	327.2974	−0.8900	44.4056
	LSTM	2.7162	16.1524	20.9142	437.4030	0.9063	46.8531	3.3677	3.1931	4.2601	18.1483	0.8952	53.8462
	PROPOSED-CM	1.0761	1.0133	1.3733	1.8860	0.9891	85.0174	2.5337	2.3974	3.2848	10.7896	0.9377	53.1469
Brent	ARIMA	2.0583	2.0215	2.8544	8.1474	0.9514	56.7944	4.1154	4.0474	5.5389	30.6791	0.8168	55.5944
	BPNN	6.7429	7.1465	9.9401	98.8055	0.4109	48.0836	8.8325	9.2905	12.5022	156.3061	0.0669	46.8531
	ELMAN	5.7760	5.9586	7.7749	60.4488	0.6396	46.3415	10.1002	10.5557	14.0505	197.4172	−0.1786	48.2517
	GRNN	13.0120	13.6462	17.4663	305.0732	−0.8188	47.3868	15.4317	16.0845	19.9941	399.7633	−1.3866	46.1538
	ELMNN	13.4468	14.5495	26.3436	693.9873	−3.1375	49.8258	11.0481	11.7554	16.7555	280.7464	−0.6760	47.9021
	SVM	2.1524	2.1177	3.0080	9.0478	0.9461	46.3415	3.0638	3.0199	4.1824	17.4925	0.8956	53.4965
	FTDNN	8.5305	9.1614	21.2842	453.0175	−1.7008	51.5679	62.4609	62.9054	71.3878	5096.2212	−29.4240	48.6014
	NAR	5.1948	5.4136	8.0547	64.8778	0.6132	46.3415	16.1820	17.0227	26.5758	706.2733	−3.2164	55.2448
	LSTM	2.4174	2.3983	3.2836	10.7818	0.9357	51.2195	4.7115	4.7569	6.1100	37.3320	0.7771	47.9021
	PROPOSED-CM	1.1104	1.0973	1.5266	2.3305	0.9861	80.8362	1.6290	1.6052	2.2163	4.9118	0.9707	69.2308

Note: Bolded numerical results are the values of the metrics for the out-of-sample point prediction performance of the hybrid prediction model with optimal performance on the corresponding metrics. Among them, the error accuracy indicators **MAPE**, **MAE**, **RMSE**, and **MSE** are negative, i.e., the smaller the value of the corresponding indicators, the better the performance, on the contrary, the trend accuracy and directional accuracy indicators **R²** and **DACC** are positive indicators, i.e., the larger the value of the corresponding indicators, the better the performance.

Table 5
Out-of-sample performance of the classical benchmark model versus the proposed hybrid model in **Oman** and **SC**.

		Step-1						Step-2					
		MAPE	MAE	RMSE	MSE	R ²	DACC	MAPE	MAE	RMSE	MSE	R ²	DACC
Oman	ARIMA	1.7909	1.5139	1.9963	3.9852	0.9412	55.0523	3.4848	2.9365	3.6733	13.4934	0.7996	49.3007
	BPNN	1.9793	1.6722	2.2173	4.9163	0.9274	53.3101	2.7845	2.3524	3.0832	9.5061	0.8588	49.6503
	ELMAN	4.9856	4.2006	4.9955	24.9550	0.6317	52.6132	6.9044	5.7657	7.2974	53.2515	0.2093	50.6993
	GRNN	3.0456	2.6238	3.6340	13.2062	0.8051	48.4321	3.7236	3.1890	4.3479	18.9043	0.7193	47.9021
	ELMNN	1.9307	1.6351	2.2062	4.8673	0.9282	51.2195	2.6298	2.2126	2.8777	8.2810	0.8770	47.5524
	SVM	1.8215	1.5428	2.0151	4.0607	0.9401	50.1742	2.5574	2.1544	2.7370	7.4909	0.8888	49.6503
	FTDNN	2.2679	1.9551	2.9439	8.6664	0.8721	53.6585	2.3150	1.9986	2.9583	8.7518	0.8700	50.6993
	NAR	2.1331	1.8174	2.4856	6.1784	0.9088	48.0836	3.0108	2.5647	3.4023	11.5754	0.8281	49.3007
	LSTM	2.0143	1.7033	2.2050	4.8622	0.9282	48.7805	4.2385	3.6523	4.8926	23.9375	0.6445	48.6014
		PROPOSED-CM	0.8763	0.7408	0.9769	0.9542	0.9859	1.2229	1.0323	1.3155	1.7307	0.9743	75.5245
SC	ARIMA	1.6719	9.9558	13.2073	174.4333	0.9626	53.6585	3.6372	21.6037	26.8400	720.3851	0.8457	44.4056
	BPNN	1.8107	10.7578	14.4739	209.4949	0.9551	52.6132	2.6996	15.9455	20.8864	436.2430	0.9066	50.3497
	ELMAN	2.3787	14.1769	18.6321	347.1542	0.9256	47.0383	6.5166	39.2724	48.8561	2386.9226	0.4888	47.2028
	GRNN	3.1272	18.6756	23.9144	571.8967	0.8775	49.4774	3.9779	23.6585	30.0419	902.5186	0.8067	49.3007
	ELMNN	1.7291	10.2626	13.8407	191.5653	0.9590	54.3554	2.5096	14.8414	19.3718	375.2665	0.9196	50.6993
	SVM	1.7174	10.2769	13.5672	184.0688	0.9606	47.0383	2.4447	14.5506	18.5252	343.1812	0.9265	47.5524
	FTDNN	2.7910	16.9212	22.7463	517.3940	0.8892	49.1289	3.4821	21.7136	52.0427	2708.4404	0.4199	49.3007
	NAR	2.0907	12.4180	15.7357	247.6115	0.9470	51.2195	3.0978	18.5393	23.3689	546.1058	0.8830	52.0979
	LSTM	2.0000	11.9480	16.0683	258.1913	0.9447	48.4321	2.7162	16.1524	20.9142	437.4030	0.9063	46.8531
		PROPOSED-CM	0.7803	4.6718	6.2590	39.1751	0.9916	1.0587	6.3115	8.2022	67.2761	0.9856	76.5734

Note: Bolded numerical results are the values of the metrics for the out-of-sample point prediction performance of the hybrid prediction model with optimal performance on the corresponding metrics. Among them, the error accuracy indicators **MAPE**, **MAE**, **RMSE**, and **MSE** are negative, i.e., the smaller the value of the corresponding indicators, the better the performance, on the contrary, the trend accuracy and directional accuracy indicators **R²** and **DACC** are positive indicators, i.e., the larger the value of the corresponding indicators, the better the performance.

Table 6

Out-of-sample performance comparison of hybrid prediction systems based on different denoising algorithms.

		Step-1						Step-2					
		MAPE	MAE	RMSE	MSE	R ²	DACC	MAPE	MAE	RMSE	MSE	R ²	DACC
WTI	RAW-CM	2.1430	2.0154	2.8630	8.1965	0.9527	59.5819	3.3628	3.1904	4.2520	18.0792	0.8956	54.1958
	EMD-CM	1.5536	1.4572	1.8621	3.4675	0.9800	72.1254	3.7401	3.6156	4.5943	21.1077	0.8781	49.6503
	CEEMD-CM	1.5744	1.4711	1.9759	3.9041	0.9775	72.4739	3.8133	3.6467	4.9836	24.8359	0.8566	51.7483
	SSA-CM	1.8045	1.6931	2.1900	4.7961	0.9723	69.6864	2.6656	2.5802	3.4258	11.7360	0.9322	59.4406
	VMD-CM	1.1638	1.1008	1.4755	2.1772	0.9874	80.4878	3.9725	3.7987	4.9736	24.7363	0.8572	54.5455
	PROPOSED-CM	1.0761	1.0133	1.3733	1.8860	0.9891	85.0174	2.5337	2.3974	3.2848	10.7896	0.9377	53.1469
Brent	RAW-CM	2.1131	2.0771	2.8745	8.2626	0.9507	53.3101	4.2233	4.1612	5.6192	31.5754	0.8115	53.4965
	EMD-CM	1.3319	1.3154	1.7363	3.0148	0.9820	72.8223	3.5002	3.4530	4.4248	19.5788	0.8831	49.6503
	CEEMD-CM	1.7817	1.7387	2.2221	4.9379	0.9706	63.7631	3.0164	2.9627	4.1213	16.9848	0.8986	50.0000
	SSA-CM	1.7808	1.7526	2.3520	5.5321	0.9670	65.8537	2.8941	2.8683	3.7830	14.3115	0.9146	55.5944
	VMD-CM	1.1521	1.1417	1.5710	2.4681	0.9853	77.3519	2.8941	2.8683	3.7830	14.3115	0.9146	55.5944
	PROPOSED-CM	1.1104	1.0973	1.5266	2.3305	0.9861	80.8362	1.6290	1.6052	2.2163	4.9118	0.9707	69.2308
Oman	RAW-CM	1.8120	1.5329	2.0154	4.0617	0.9401	53.6585	2.3075	1.9914	2.9427	8.6592	0.8714	51.3986
	EMD-CM	1.1415	0.9592	1.2017	1.4441	0.9787	81.1847	1.8679	1.5743	2.0499	4.2022	0.9376	65.3846
	CEEMD-CM	1.2284	1.0381	1.3556	1.8377	0.9729	72.8223	2.1223	1.7894	2.3110	5.3408	0.9207	52.4476
	SSA-CM	1.5978	1.3553	1.7697	3.1318	0.9538	69.6864	1.8154	1.5434	2.0017	4.0069	0.9405	67.1329
	VMD-CM	0.9012	0.7659	0.9905	0.9811	0.9855	82.5784	1.2346	1.0442	1.3537	1.8326	0.9728	74.1259
	PROPOSED-CM	0.8763	0.7408	0.9769	0.9542	0.9859	84.6690	1.2229	1.0323	1.3155	1.7307	0.9743	75.5245
SC	RAW-CM	1.7120	10.2115	13.3590	178.4617	0.9618	49.4774	2.4573	14.6471	19.1732	367.6120	0.9213	50.3497
	EMD-CM	0.9803	5.8966	7.3621	54.2002	0.9884	81.5331	1.6545	9.9416	12.8264	164.5154	0.9648	61.1888
	CEEMD-CM	1.2840	7.6337	9.5368	90.9506	0.9805	69.6864	2.4292	14.3398	19.0617	363.3486	0.9222	49.3007
	SSA-CM	1.4473	8.6155	11.0223	121.4920	0.9740	72.8223	1.6954	10.1084	12.8439	164.9660	0.9647	67.4825
	VMD-CM	0.7790	4.6705	6.2552	39.1275	0.9916	85.3659	1.6954	10.1084	12.8439	164.9660	0.9647	67.4825
	PROPOSED-CM	0.7803	4.6718	6.2590	39.1751	0.9916	84.6690	1.0587	6.3115	8.2022	67.2761	0.9856	76.5734

Note: Bolded numerical results are the values of the metrics for the out-of-sample point prediction performance of the hybrid prediction model with optimal performance on the corresponding metrics. Among them, the error accuracy indicators MAPE, MAE, RMSE, and MSE are negative, i.e., the smaller the value of the corresponding indicators, the better the performance, on the contrary, the trend accuracy and directional accuracy indicators R² and DACC are positive indicators, i.e., the larger the value of the corresponding indicators, the better the performance. In addition, RAW-CM, EMD-CM, CEEMD-CM, SSA-CM, and VMD-CM are hybrid point prediction models constructed based on the original data, EMD-based denoised data, CEEMD-based denoised data, SSA-based denoised data, and VMD-based denoised data, respectively.

Table 7

Out-of-sample performance comparison of fuzzy interval forecasting models based on different denoising algorithms.

		Step-1				Step-2			
		FICP	FINAW	AIS	CWC	FICP	FINAW	AIS	CWC
WTI	RAW-CM-FIP	96.5278	0.2372	−1.6122	0.2372	96.1672	0.3423	−2.3520	0.3423
	EMD-CM-FIP	95.4861	0.1423	−0.9422	0.1423	96.1672	0.3175	−2.1440	0.3175
	CEEMD-CM-FIP	95.1389	0.1430	−1.0718	0.1430	95.4704	0.3587	−2.7905	0.3587
	SSA-CM-FIP	95.8333	0.1647	−1.1630	0.1647	95.8188	0.2504	−1.7562	0.2504
	VMD-CM-FIP	95.8333	0.1137	−0.8013	0.1137	95.1219	0.3191	−2.4480	0.3191
	PROPOSED-CM-FIP	95.8333	0.0973	−0.7012	0.0973	95.4704	0.2333	−1.6307	0.2333
Brent	RAW-CM-FIP	95.4861	0.2210	−1.5921	0.2210	95.1219	0.4159	−3.6211	0.4159
	EMD-CM-FIP	96.8750	0.1431	−0.9689	0.1431	95.1219	0.3043	−2.3490	0.3043
	CEEMD-CM-FIP	95.1389	0.1628	−1.1668	0.1628	95.8188	0.2781	−1.9968	0.2781
	SSA-CM-FIP	95.4861	0.1706	−1.2003	0.1706	95.1219	0.2600	−2.0892	0.2600
	VMD-CM-FIP	95.8333	0.1217	−0.8766	0.1217	95.1219	0.2074	−1.5725	0.2074
	PROPOSED-CM-FIP	96.1806	0.1165	−0.7868	0.1165	95.1219	0.1638	−1.1667	0.1638
Oman	RAW-CM-FIP	96.8750	0.2359	−1.0844	0.2359	95.4704	0.3171	−1.3797	0.3171
	EMD-CM-FIP	95.1389	0.1546	−0.6422	0.1546	95.4704	0.2349	−1.0516	0.2349
	CEEMD-CM-FIP	95.4861	0.1590	−0.7408	0.1590	95.4704	0.2746	−1.2703	0.2746
	SSA-CM-FIP	96.1806	0.2236	−0.9737	0.2236	95.1219	0.2422	−1.0971	0.2422
	VMD-CM-FIP	95.8333	0.1278	−0.5236	0.1278	96.1672	0.1801	−0.7658	0.1801
	PROPOSED-CM-FIP	96.5278	0.1312	−0.5332	0.1312	96.5157	0.1626	−0.6832	0.1626
SC	RAW-CM-FIP	95.8333	0.2510	−8.0978	0.2510	95.8188	0.3368	−10.7235	0.3368
	EMD-CM-FIP	95.4861	0.1288	−3.9897	0.1288	96.5157	0.2185	−6.9901	0.2185
	CEEMD-CM-FIP	95.4861	0.1718	−5.8376	0.1718	95.1219	0.3263	−10.7267	0.3263
	SSA-CM-FIP	95.1389	0.2389	−6.9614	0.2389	95.1219	0.2511	−7.5595	0.2511
	VMD-CM-FIP	95.8333	0.1121	−3.6876	0.1121	95.8188	0.1696	−5.7872	0.1696
	PROPOSED-CM-FIP	95.4861	0.1113	−3.6437	0.1113	96.8641	0.1617	−5.3501	0.1617

Note: The numerical results in bold are the values of the corresponding metrics for the hybrid interval prediction model that has the best performance in the corresponding metrics. Among them, **FICP** and **AIS** are positive indicators, i.e., the larger the value of the indicator the better the corresponding performance, and **FINAW**, **AWD**, and **CWC** are negative indicators, i.e., the smaller the value of the indicator means the better the corresponding performance of the model. In addition, **RAW-CM-FIP**, **EMD-CM-FIP**, **CEEMD-CM-FIP**, **SSA-CM-FIP**, and **VMD-CM-FIP** are hybrid interval prediction models constructed based on the original data, **EMD**-based denoised data, **CEEMD**-based denoised data, **SSA**-based denoised data, and **VMD**-based denoised data, respectively.

$CWC_{\text{PROPOSED-CM-FIP}}^{\text{Oman}} = 0.1626$, $FICP_{\text{PROPOSED-CM-FIP}}^{\text{SC}} = 96.8641\%$, $FINAW_{\text{PROPOSED-CM-FIP}}^{\text{SC}} = 0.1617$, $AIS_{\text{PROPOSED-CM-FIP}}^{\text{Oman}} = -5.3501$, $CWC_{\text{PROPOSED-CM-FIP}}^{\text{Oman}} = 0.1617$, and the performance improvement percentage is within [4 %,52 %]. As for **WTI** and **Brent**, compared to other fuzzy interval forecasting models, **PROPOSED-CM-FIP** achieves the optimal performance in all performance metrics except for **FICP**, with the percentage of performance improvement in the [6 %,40 %] and [20 %,68 %] intervals, respectively.

4. Discussion

In this section, depending on the empirical findings of the previous section, we establish the ratio metrics system and the Diebold-Mariano (**DM**) test framework to carry out further exploration of the model's performance advantages.

Discussion 1: Natural benchmarks and performance improvement.

Since **CO** price forecasting is a challenging task and many studies have concluded that no forecasting strategy can beat the stochastic wandering strategy, thus, this study introduces the stochastic wandering model as the natural benchmark model to implement further performance discussion. The 1- and 2-step-ahead numerical simulation results of the four **CO** futures price series from the random walk model are presented in Table 8, while Tables 9 and 10 take the random walk model as the natural benchmark to further illustrate the performance improvement of the models presented in Tables 4–6 in comparison with the natural benchmark model by constructing ratio metrics. Specifically, the ratio metrics involved are as shown in Equations (20) and (21):

$$\begin{aligned}
 R_{\text{MAPE}} &= \text{MAPE}^i / \text{MAPE}^{r-w}, \\
 R_{\text{MAE}} &= \text{MAE}^i / \text{MAE}^{r-w}, \\
 R_{\text{RMSE}} &= \text{RMSE}^i / \text{RMSE}^{r-w}, \\
 R_{\text{MSE}} &= \text{MSE}^i / \text{MSE}^{r-w}.
 \end{aligned} \tag{20}$$

$$\begin{aligned}
 R_{R^2} &= R^{2r-w} / R^{2i}, \\
 R_{\text{DACC}} &= \text{DACC}^{r-w} / \text{DACC}^i.
 \end{aligned} \tag{21}$$

Where superscript i denotes the i -th model used for comparison, including (**ARIMA**, **BPNN**, **ELMAN**, **GRNN**, **ELMNN**, **SVM**, **FTDNN**, **NAR**, **LSTM**, **RAW-CM**, **EMD-CM**, **CEEMD-CM**, **SSA-CM**, **VMD-CM**, **PROPOSED-CM**), and superscript r_w refers to the natural benchmark model. In particular, since the two metrics, R^2 and **DACC**, are positive metrics, and the rise in their values implies an improvement in performance, so we take the inverse of them to ensure that all ratio metrics in Equations. (20) and (21) within the (0,1)

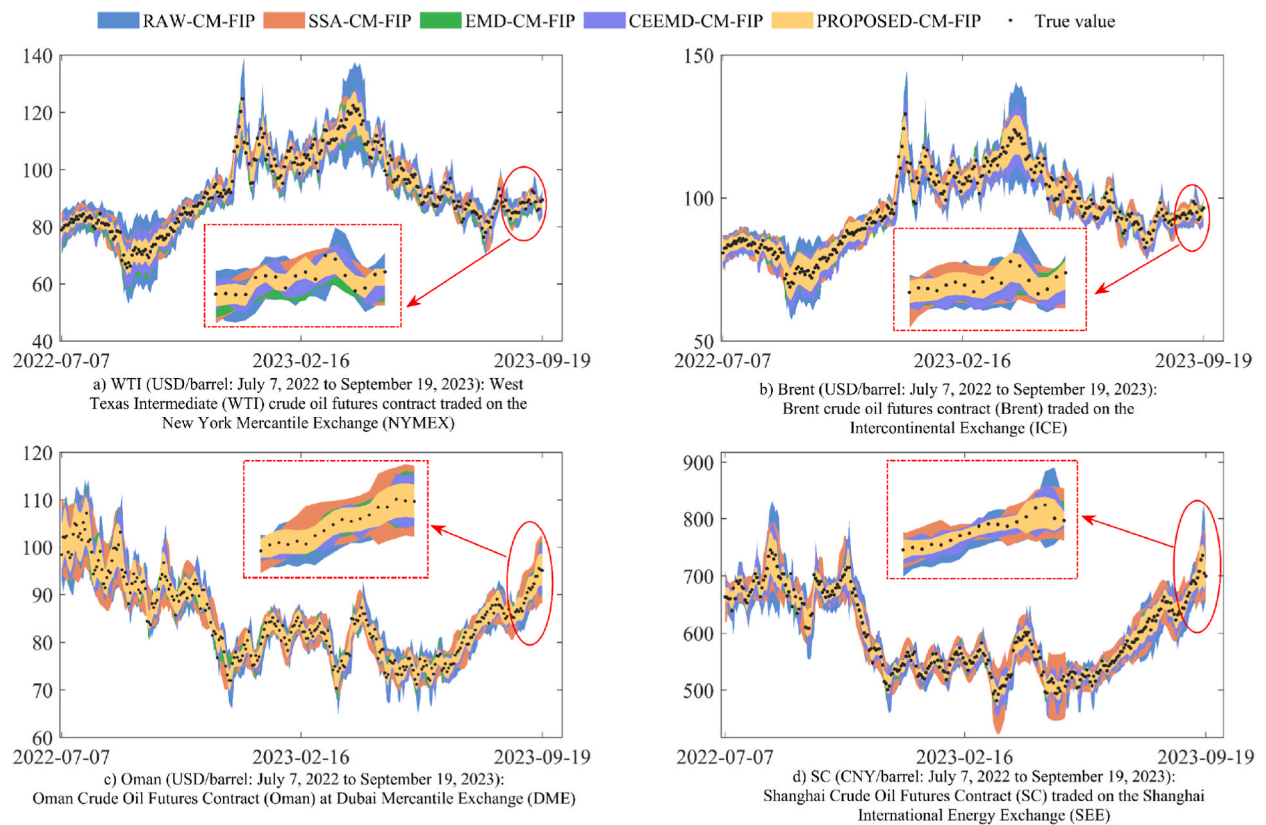


Fig. 5. Schematic of one-step-ahead out-of-sample interval forecasting results.

Table 8

Results of the stochastic wandering model in four crude oil futures markets.

	Step-1				Step-2			
	WTI	Brent	Oman	SC	WTI	Brent	Oman	SC
MAPE	2.2103	2.1154	1.7911	1.6868	3.2320	3.0266	2.5672	2.4319
MAE	2.0798	2.0810	1.5163	10.0632	3.0455	2.9778	2.1680	14.4934
RMSE	2.9427	2.9365	2.0030	13.4605	4.1870	4.1798	2.7767	18.4105
MSE	8.6593	8.6230	4.0121	181.1847	17.5312	17.4707	7.7103	338.9459
R ²	0.9500	0.9486	0.9408	0.9612	0.8988	0.8958	0.8862	0.9274
DACC	100.0000	100.0000	100.0000	100.0000	52.9617	53.3101	49.1289	46.6899

indicate that the model outperforms the natural benchmark model, while those ratio values outside the (0,1) indicate that the model underperforms the natural benchmark model.

The results in Table 8 through 10 show us that in most cases, the single forecasting models constructed on raw and unprocessed data are unable to beat the stochastic wandering model in terms of performance, which is a similar finding to that of previous studies, and such a result may be attributed to the fact that the crude oil price series were not very volatile during the validation period. However, after a series of improvement strategies including denoising, screening, and optimal integration, the hybrid forecasting system proposed in this study is able to significantly outperform the stochastic wandering model, which is the natural benchmark, in all performance metrics except for the DACC in 1-step-ahead forecasting.

Discussion 2: Statistical significance testing.

Moreover, we introduce the DM test to verify whether the performance improvement of the constructed forecasting model is statistically significant. Theoretically, this test targets the forecast error and constructs a test statistic by two-by-two comparison to test whether it obeys a normal distribution. The test results in Table 11 show that the performance advantage of our proposed hybrid forecasting system over other models is significant on all datasets and not due to chance.

Table 9
Model performance improvement compared to the natural benchmark in **WTI** and **Brent**.

		Step-1						Step-2					
		R_{MAPE}	R_{MAE}	R_{RMSE}	R_{MSE}	R_{R^2}	R_{DACC}	R_{MAPE}	R_{MAE}	R_{RMSE}	R_{MSE}	R_{R^2}	R_{DACC}
WTI	ARIMA	1.6455	10.3873	9.1210	83.1920	1.1233	2.2520	1.3567	1.3572	1.3344	1.7807	1.0965	0.9408
	BPNN	1.2214	7.6667	7.0978	50.3785	1.0479	1.9861	3.9088	4.2103	3.9668	15.7353	−1.5157	1.1563
	ELMAN	2.9482	18.8825	16.6026	275.6483	1.9436	2.1185	3.9001	4.1829	3.8880	15.1163	−1.6949	1.1652
	GRNN	1.7997	11.3752	10.2091	104.2253	1.1776	2.0284	4.5633	4.8929	4.5073	20.3155	−0.8506	1.1563
	ELMNN	1.1354	7.1359	6.5831	43.3368	1.0331	1.9724	1.9815	2.1366	2.1760	4.7349	1.7261	1.1220
	SVM	1.1060	6.9961	6.2954	39.6315	1.0254	2.1029	1.0329	1.0373	1.0210	1.0425	1.0048	1.0304
	FTDNN	1.5754	10.4401	17.6855	312.7781	2.2625	2.0284	2.2651	2.4319	2.3753	5.6422	2.0961	1.1563
	NAR	1.4015	8.9139	7.9414	63.0658	1.0759	1.9195	4.3960	4.6957	4.3208	18.6694	−1.0099	1.1927
	LSTM	1.2289	7.7662	7.1072	50.5125	1.0482	2.1343	1.0420	1.0485	1.0175	1.0352	1.0040	0.9836
	RAW-CM	0.9695	0.9690	0.9729	0.9466	0.9972	1.6784	1.0405	1.0476	1.0155	1.0313	1.0036	0.9772
	EMD-CM	0.7029	0.7006	0.6328	0.4004	0.9694	1.3865	1.1572	1.1872	1.0973	1.2040	1.0236	1.0667
	CEEMD-CM	0.7123	0.7073	0.6715	0.4509	0.9719	1.3798	1.1799	1.1974	1.1902	1.4167	1.0493	1.0234
	SSA-CM	0.8164	0.8141	0.7442	0.5539	0.9771	1.4350	0.8247	0.8472	0.8182	0.6694	0.9642	0.8910
	VMD-CM	0.5265	0.5293	0.5014	0.2514	0.9621	1.2424	1.2291	1.2473	1.1879	1.4110	1.0485	0.9710
	PROPOSED-CM	0.4868	0.4872	0.4667	0.2178	0.9605	1.1762	0.7839	0.7872	0.7845	0.6155	0.9585	0.9965
Brent	ARIMA	0.9730	0.9714	0.9720	0.9448	0.9970	1.7607	1.3597	1.3592	1.3252	1.7560	1.0968	0.9589
	BPNN	3.1875	3.4342	3.3850	11.4584	2.3086	2.0797	2.9183	3.1199	2.9911	8.9468	13.3908	1.1378
	ELMAN	2.7305	2.8633	2.6477	7.0102	1.4831	2.1579	3.3372	3.5448	3.3615	11.2999	−5.0159	1.1048
	GRNN	6.1511	6.5575	5.9480	35.3790	−1.1585	2.1103	5.0987	5.4015	4.7835	22.8820	−0.6461	1.1551
	ELMNN	6.3566	6.9916	8.9711	80.4810	−0.3023	2.0070	3.6503	3.9477	4.0087	16.0696	−1.3252	1.1129
	SVM	1.0175	1.0176	1.0244	1.0493	1.0026	2.1579	1.0123	1.0141	1.0006	1.0013	1.0003	0.9965
	FTDNN	4.0326	4.4024	7.2482	52.5360	−0.5577	1.9392	20.6374	21.1249	17.0793	291.7018	−0.0304	1.0969
	NAR	2.4557	2.6014	2.7430	7.5238	1.5470	2.1579	5.3466	5.7166	6.3582	40.4263	−0.2785	0.9650
	LSTM	1.1428	1.1525	1.1182	1.2504	1.0138	1.9524	1.5567	1.5975	1.4618	2.1368	1.1528	1.1129
	RAW-CM	0.9989	0.9981	0.9789	0.9582	0.9978	1.8758	1.3954	1.3974	1.3444	1.8073	1.1039	0.9965
	EMD-CM	0.6296	0.6321	0.5913	0.3496	0.9660	1.3732	1.1565	1.1596	1.0586	1.1207	1.0144	1.0737
	CEEMD-CM	0.8423	0.8355	0.7567	0.5726	0.9773	1.5683	0.9966	0.9949	0.9860	0.9722	0.9969	1.0662
	SSA-CM	0.8418	0.8422	0.8010	0.6416	0.9810	1.5185	0.9562	0.9632	0.9051	0.8192	0.9795	0.9589
	VMD-CM	0.5446	0.5486	0.5350	0.2862	0.9627	1.2928	0.9562	0.9632	0.9051	0.8192	0.9795	0.9589
	PROPOSED-CM	0.5249	0.5273	0.5199	0.2703	0.9620	1.2371	0.5382	0.5391	0.5302	0.2811	0.9229	0.7700

Note: Values of metrics falling within (0,1) that are bolded indicate that the corresponding model outperforms the natural benchmark model.

Table 10
Model performance improvement compared to the natural benchmark in **Oman** and **SC**.

		Step-1						Step-2					
		R_{MAPE}	R_{MAE}	R_{RMSE}	R_{MSE}	R_{R^2}	R_{DACC}	R_{MAPE}	R_{MAE}	R_{RMSE}	R_{MSE}	R_{R^2}	R_{DACC}
Oman	ARIMA	0.9999	0.9984	0.9966	0.9933	0.9996	1.8165	1.3574	1.3545	1.3229	1.7501	1.1083	0.9965
	BPNN	1.1051	1.1028	1.1070	1.2254	1.0144	1.8758	1.0846	1.0850	1.1104	1.2329	1.0319	0.9895
	ELMAN	2.7835	2.7703	2.4940	6.2199	1.4893	1.9007	2.6895	2.6594	2.6280	6.9066	4.2341	0.9690
	GRNN	1.7004	1.7304	1.8142	3.2916	1.1685	2.0647	1.4504	1.4709	1.5658	2.4518	1.2320	1.0256
	ELMNN	1.0779	1.0784	1.1014	1.2131	1.0136	1.9524	1.0244	1.0206	1.0364	1.0740	1.0105	1.0332
	SVM	1.0170	1.0175	1.0060	1.0121	1.0007	1.9931	0.9962	0.9937	0.9857	0.9715	0.9971	0.9895
	FTDNN	1.2662	1.2894	1.4697	2.1600	1.0788	1.8636	0.9018	0.9219	1.0654	1.1351	1.0186	0.9690
	NAR	1.1909	1.1986	1.2409	1.5399	1.0352	2.0797	1.1728	1.1830	1.2253	1.5013	1.0702	0.9965
	LSTM	1.1246	1.1233	1.1008	1.2119	1.0136	2.0500	1.6510	1.6846	1.7620	3.1046	1.3750	1.0109
	RAW-CM	1.0117	1.0110	1.0062	1.0124	1.0007	1.8636	0.8988	0.9185	1.0598	1.1231	1.0170	0.9558
	EMD-CM	0.6373	0.6326	0.5999	0.3599	0.9613	1.2318	0.7276	0.7261	0.7382	0.5450	0.9452	0.7514
	CEEMD-CM	0.6858	0.6846	0.6768	0.4580	0.9670	1.3732	0.8267	0.8254	0.8323	0.6927	0.9625	0.9367
	SSA-CM	0.8921	0.8938	0.8835	0.7806	0.9864	1.4350	0.7071	0.7119	0.7209	0.5197	0.9423	0.7318
SC	VMD-CM	0.5032	0.5051	0.4945	0.2445	0.9546	1.2110	0.4809	0.4816	0.4875	0.2377	0.9110	0.6628
	PROPOSED-CM	0.4892	0.4886	0.4877	0.2378	0.9542	1.1811	0.4764	0.4761	0.4738	0.2245	0.9096	0.6505
	ARIMA	0.9912	0.9893	0.9812	0.9627	0.9985	1.8636	1.4956	1.4906	1.4579	2.1254	1.0966	1.0514
	BPNN	1.0734	1.0690	1.0753	1.1563	1.0064	1.9007	1.1101	1.1002	1.1345	1.2871	1.0229	0.9273
	ELMAN	1.4102	1.4088	1.3842	1.9160	1.0384	2.1259	2.6796	2.7097	2.6537	7.0422	1.8973	0.9891
	GRNN	1.8539	1.8558	1.7766	3.1564	1.0954	2.0211	1.6357	1.6324	1.6318	2.6627	1.1496	0.9470
	ELMNN	1.0251	1.0198	1.0282	1.0573	1.0023	1.8397	1.0320	1.0240	1.0522	1.1072	1.0085	0.9209
	SVM	1.0181	1.0212	1.0079	1.0159	1.0006	2.1259	1.0053	1.0039	1.0062	1.0125	1.0010	0.9819
	FTDNN	1.6546	1.6815	1.6899	2.8556	1.0810	2.0355	1.4318	1.4982	2.8268	7.9908	2.2086	0.9470
	NAR	1.2394	1.2340	1.1690	1.3666	1.0150	1.9524	1.2738	1.2792	1.2693	1.6112	1.0503	0.8962
	LSTM	1.1857	1.1873	1.1937	1.4250	1.0174	2.0647	1.1169	1.1145	1.1360	1.2905	1.0233	0.9965
	RAW-CM	1.0149	1.0147	0.9925	0.9850	0.9994	2.0211	1.0104	1.0106	1.0414	1.0846	1.0066	0.9273
	EMD-CM	0.5812	0.5860	0.5469	0.2991	0.9725	1.2265	0.6803	0.6859	0.6967	0.4854	0.9612	0.7630
	CEEMD-CM	0.7612	0.7586	0.7085	0.5020	0.9803	1.4350	0.9989	0.9894	1.0354	1.0720	1.0056	0.9470
	SSA-CM	0.8580	0.8561	0.8189	0.6705	0.9868	1.3732	0.6972	0.6974	0.6976	0.4867	0.9613	0.6919
	VMD-CM	0.4618	0.4641	0.4647	0.2160	0.9693	1.1714	0.6972	0.6974	0.6976	0.4867	0.9613	0.6919
	PROPOSED-CM	0.4626	0.4642	0.4650	0.2162	0.9693	1.1811	0.4353	0.4355	0.4455	0.1985	0.9409	0.6097

Note: Values of metrics falling within (0,1) that are bolded indicate that the corresponding model outperforms the natural benchmark model.

5. Conclusion

Sharp fluctuations in CO prices may exacerbate global economic crises, political crises, and financial market crises; therefore, accurate and reliable CO price forecasting is a shared global goal. This study is dedicated to constructing a hybrid CO price forecasting system that can simultaneously perform deterministic forecasting and uncertainty analysis, aiming to provide CO market participants and managers with a comprehensive hint of expected risks from the dual perspectives of point forecasting and interval forecasting. The system takes “denoising-integration” as the core of the research, firstly, it integrates multiple denoising techniques and significantly reduces the impact of noise on the forecasting results based on synthesizing the performance advantages of various denoising techniques. Secondly, in the process of deterministic forecasting, a multi-objective intelligent optimization algorithm is introduced to optimize the model weights of multiple sub-forecasting models integrated with denoising techniques, so as to establish a robust and adaptive hybrid forecasting model, which significantly improves the error accuracy, trend accuracy, and directional accuracy of the deterministic forecasting of CO prices, and reduces the forecasting uncertainty caused by the model uncertainty and data uncertainty. Finally, on the foundation of deterministic forecasting, fuzzy theory is incorporated to establish a fuzzy interval forecasting model with the minimum interval width and optimal comprehensive performance by using a multi-objective optimization algorithm on the premise of ensuring that the coverage of forecasting intervals meets the requirements of the confidence level. The empirical study confirms that the hybrid forecasting system constructed in this study achieves excellent forecasting performance in the four core CO futures markets of interest, and can provide valuable reference information for stakeholders to formulate relevant policies and manage expected risks.

CRedit authorship contribution statement

Hua Luo: Writing – review & editing, Visualization, Validation, Investigation, Funding acquisition, Formal analysis. **Yue Yu:** Writing – original draft, Methodology, Conceptualization.

Statements and Declarations

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Data and code availability statement

Data included in article is referenced in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 11
Statistical significance results from DM test.

Model	DME		ICE		INE		NYMAX		Average	
	Step-1	Step-2	Step-1	Step-2	Step-1	Step-2	Step-1	Step-2	Step-1	Step-2
Random walk	^a 7.6570	^a 6.6910	^a 5.2771	^a 7.1473	^a 6.6599	^a 10.808	^a 5.7776	^a 6.5098	^a 6.3429	^a 7.7892
ARIMA	^a 7.7377	^a 10.662	^a 5.3644	^a 11.614	^a 7.0346	^a 9.5247	^a 5.8318	^a 12.106	^a 6.4921	^a 10.976
BPNN	^a 7.7718	^a 8.1716	^a 10.526	^a 10.092	^a 8.2882	^a 12.811	^a 11.873	^a 12.165	^a 9.6148	^a 10.810
ELMAN	^a 14.617	^a 12.267	^a 11.832	^a 14.494	^a 9.4286	^a 9.5180	^a 7.7564	^a 13.017	^a 10.908	^a 12.324
GRNN	^a 7.2117	^a 7.5525	^a 13.616	^a 9.8626	^a 9.2788	^a 9.7217	^a 14.120	^a 8.3339	^a 11.057	^a 8.8677
ELMNN	^a 7.7872	^a 9.1636	^a 6.3350	^a 6.7025	^a 7.5371	^a 9.7402	^a 5.1251	^a 5.0308	^a 6.6961	^a 7.6593
SVM	^a 8.0657	^a 9.5351	^a 5.2339	^a 28.750	^a 6.9810	^a 2.9422	^a 5.6528	^a 10.253	^a 6.4834	^a 12.870
FTDNN	^a 5.1671	^a 5.3168	^a 2.9080	^a 9.1370	^a 9.3168	^a 11.300	^a 10.312	^a 12.775	^a 6.9262	^a 9.6323
NAR	^a 6.7840	^a 7.9477	^a 5.5991	^a 9.7634	^a 9.7405	^a 9.3176	^a 10.979	^a 5.5108	^a 8.2757	^a 8.1349
LSTM	^a 8.0694	^a 9.1683	^a 7.8131	^a 4.2242	^a 7.5575	^a 5.9738	^a 9.0582	^b 2.272	^a 8.1246	^a 4.2734
RAW-CM	^a 7.8486	^a 5.3238	^a 5.6575	^a 7.2580	^a 7.1603	^a 9.6195	^a 5.8272	^a 5.5049	^a 6.6234	^a 6.9265
EMD-CM	^a 4.2288	^a 6.9086	^a 2.7216	^a 9.5897	^a 3.4647	^a 4.6467	^a 4.6327	^a 6.6511	^a 3.7619	^a 6.9490
CEEMD-CM	^a 5.0735	^a 8.4503	^a 6.4187	^a 6.7466	^a 6.5484	^a 7.5494	^a 4.2221	^a 6.0100	^a 5.5657	^a 7.1891
SSA-CM	^a 7.9490	^a 6.7526	^a 5.9531	^a 6.8492	^a 8.4023	^a 8.7357	^a 5.6077	0.7000	^a 6.9780	^a 5.7593
VMD-CM	^a 5.0735	^a 8.4503	^a 6.4187	^a 6.7466	^a 6.5484	^a 7.5494	^a 4.2221	^a 6.0100	^a 5.5657	^a 7.1891

Note: Levels of significance.

^a for 1 %.

^b for 5 %, * for 10 %.

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